2nd model

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0.1 Model 2: Realistic Capacity Expansion and Location

Model 2 builds on Model 1 and retains the same desert-elimination requirement, the same under-5 policy target, and the same basic facility-type structure. Therefore, instead of repeating the common notation, this section focuses only on the modifications introduced in the more realistic setting.

0.1.1 Additional Modeling Features

Relative to Model 1, Model 2 introduces three main changes. First, expansion is modeled at the *individual facility level* rather than being aggregated at the zip-code level. Second, the expansion cost is no longer linear; instead, it is piecewise increasing depending on the expansion ratio of each facility. Third, new facilities are selected from explicit candidate locations and must satisfy a same-zip minimum-distance rule.

0.1.2 Additional Sets, Parameters, and Decision Variables

Let

$$F = \text{set of existing facilities}, \qquad L = \text{set of candidate locations for new facilities}.$$

For each existing facility $f \in F$, let:

- n_f: current total capacity,
- $z(f)$: zip code of facility f.

For each candidate location $l \in L$, let:

- $z(l)$: zip code of location l.

For each facility type $s \in \mathcal{S} = \{1, 2, 3\}$, the number of total slots and under-5 slots remain the same as in Model 1. However, in Model 2, the construction cost parameter already includes the additional \$100 per under-5 slot for specialized equipment. Therefore, the costs used in the implementation are:

$$C_1 = 70000, \qquad C_2 = 105000, \qquad C_3 = 135000.$$

The decision variables are:

- $x_f \geq 0$: total expansion slots added at existing facility f;
- $x_f^{(u5)} \geq 0$: under-5 expansion slots added at facility f;
- $y_{ls} \in \{0, 1\}$: equals 1 if a new type-s facility is built at candidate location l, and 0 otherwise;
- $c_f^{\text{exp}} \geq 0$: total expansion cost associated with facility f.

To model the piecewise cost structure, define binary variables:

$$b_{f1}, b_{f2}, b_{f3} \in \{0, 1\},$$

where:

- $b_{f1} = 1$ if facility f expands in the first segment $(0, 0.10n_f]$,
- $b_{f2} = 1$ if facility f expands in the second segment $[0.10n_f, 0.15n_f]$,
- $b_{f3} = 1$ if facility f expands in the third segment $[0.15n_f, 0.20n_f]$.

0.1.3 Objective Function

The objective is to minimize the total cost of expansion and new construction:

$$\min \sum_{f \in F} c_f^{\text{exp}} + \sum_{l \in L} \sum_{s \in \mathcal{S}} C_s y_{ls}.$$

Note that the construction cost C_s for each new facility type already includes the under-5 equipment cost. In contrast, for expansion of existing facilities, the under-5 equipment cost is modeled explicitly through $x_f^{(u5)}$ in the linearized expansion-cost constraints below.

0.1.4 Coverage Constraints

The total-slot and under-5 requirements remain the same in spirit as in Model 1, but they are now written using facility-level expansion and location-level construction variables. For each zip code $z \in \mathcal{Z}$,

$$S_z^{\text{curr}} + \sum_{f : z(f) = z} x_f + \sum_{l : z(l) = z} \sum_{s \in \mathcal{S}} K_s^{\text{total}} y_{ls} \geq R_z^{\text{total}},$$

$$S_z^{\text{curr},0\text{-}5} + \sum_{f : z(f) = z} x_f^{(u5)} + \sum_{l : z(l) = z} \sum_{s \in \mathcal{S}} K_s^{0\text{-}5} y_{ls} \geq R_z^{0\text{-}5}.$$

Unlike Model 1, Model 2 does not use a separate decision variable for the actual under-5 allocation of new facilities. Instead, each facility type contributes a fixed number of under-5 slots, namely $K_s^{0\text{-}5}$.

0.1.5 Facility-Level Expansion Constraints

Each existing facility can expand by at most 20% of its current capacity:

$$0 \le x_f \le 0.20 n_f, \qquad \forall f \in F.$$

The under-5 portion of expansion cannot exceed total expansion:

$$0 \le x_f^{(u5)} \le x_f, \qquad \forall f \in F.$$

0.1.6 Piecewise Segment Linkage Constraints

At most one expansion segment can be selected for each facility:

$$b_{f1} + b_{f2} + b_{f3} \le 1, \qquad \forall f \in F.$$

The following linkage constraints ensure that the value of x_f is consistent with the selected expansion segment:

$$x_f \le 0.10 n_f\, b_{f1} + 0.15 n_f\, b_{f2} + 0.20 n_f\, b_{f3}, \qquad \forall f \in F,$$

$$x_f \ge \varepsilon b_{f1} + 0.10 n_f\, b_{f2} + 0.15 n_f\, b_{f3}, \qquad \forall f \in F,$$

where $\varepsilon > 0$ is a very small constant.

Together, these constraints imply:

- if $b_{f1} = 1$, then $0 < x_f \le 0.10 n_f$;
- if $b_{f2} = 1$, then $0.10 n_f \le x_f \le 0.15 n_f$;
- if $b_{f3} = 1$, then $0.15 n_f \le x_f \le 0.20 n_f$;
- if $b_{f1} = b_{f2} = b_{f3} = 0$, then $x_f = 0$.

0.1.7 Linearized Expansion Cost Constraints

The expansion cost is modeled using an auxiliary variable c_f^{exp} and Big-M linearization. Define the slope coefficients:

$$a_{f1} = \frac{20000 + 200 n_f}{n_f}, \qquad a_{f2} = \frac{20000 + 400 n_f}{n_f}, \qquad a_{f3} = \frac{20000 + 1000 n_f}{n_f}.$$

Then for each $f \in F$,

$$c_f^{\text{exp}} \ge a_{f1} x_f + 100 x_f^{(u5)} - M_f (1 - b_{f1}),$$

$$c_f^{\text{exp}} \ge a_{f2} x_f + 100 x_f^{(u5)} - M_f (1 - b_{f2}),$$

$$c_f^{\text{exp}} \ge a_{f3} x_f + 100 x_f^{(u5)} - M_f (1 - b_{f3}),$$

$$c_f^{\text{exp}} \le M_f (b_{f1} + b_{f2} + b_{f3}),$$

where M_f is a sufficiently large constant.

Because the objective minimizes total cost, these constraints ensure that c_f^{exp} equals the appropriate expansion-cost expression for the active segment, including the additional \$100 per under-5 expansion slot.

0.1.8 New Facility Selection Constraints

At most one facility type can be chosen at each candidate location:

$$\sum_{s\in\mathcal{S}} y_{ls} \le 1, \qquad \forall l \in L.$$

0.1.9 Distance Constraints

The minimum-distance rule is enforced *only within the same zip code*, consistent with the implementation.

First, if two candidate locations l and l' in the same zip code are less than 0.06 miles apart, they cannot both be selected. Let

$$\mathcal{P}_z = \{(l,l') : z(l) = z(l') = z,\ \mathrm{dist}(l,l') < 0.06\}.$$

Then

$$\sum_{s\in\mathcal{S}} y_{ls} + \sum_{s\in\mathcal{S}} y_{l's} \le 1, \qquad \forall (l,l') \in \mathcal{P}_z,\ \forall z \in \mathcal{Z}.$$

Second, if a candidate location is within 0.06 miles of any existing facility in the same zip code, that candidate location is excluded from consideration. Let

$$\mathcal{B}_z = \{l \in L : z(l) = z,\ \exists f \in F \text{ with } z(f) = z \text{ and } \mathrm{dist}(l,f) < 0.06\}.$$

Then

$$\sum_{s\in\mathcal{S}} y_{ls} = 0, \qquad \forall l \in \mathcal{B}_z,\ \forall z \in \mathcal{Z}.$$

0.1.10 Interpretation

Model 2 is a more realistic extension of Model 1 because it replaces zip-level expansion with facility-level decisions and explicitly accounts for spatial feasibility. These changes make the model substantially more restrictive.

First, expansion can no longer be treated as a simple aggregated resource at the zip level. Each facility has its own capacity limit, and expansion becomes progressively more expensive as it moves into higher tiers. This prevents the model from relying too heavily on cheap linear expansion.

Second, new construction is no longer represented only by the number of facilities built in each zip code. Instead, each new facility must be placed at a feasible candidate location, and nearby candidate locations cannot be selected simultaneously. In addition, some candidate locations are ruled out entirely if they are too close to existing providers in the same zip code.

Third, under-5 capacity from new construction is fixed by facility type in Model 2. This differs from Model 1, where the under-5 portion of new capacity could be chosen more flexibly. As a result, Model 2 more tightly couples facility type selection with the age composition of new capacity.

Overall, these modeling choices make Model 2 a joint capacity-expansion and facility-location problem under realistic operational constraints.

0.1.11 Findings

The model was solved using Gurobi with a 600-second time limit and a 1% MIP gap. The solver returned status code 2, indicating an optimal solution. The minimum total cost is

$$\boxed{\$212,671,816.67}.$$

The solution uses both expansion of existing facilities and new construction:

- Existing facilities expanded: 1117
- New facilities built: 1513

Among the newly built facilities,

- 12 are small,
- 8 are medium,
- 1493 are large.

These results show that once realistic expansion caps, piecewise expansion costs, and spatial constraints are introduced, the model relies primarily on new large facilities rather than aggressive expansion of existing providers. Expansion is still used frequently, but mainly as a targeted supplement. Large facilities dominate because they provide the greatest total and under-5 capacity per site at the lowest average cost per slot under the given cost structure.

This pattern also suggests that the under-5 requirement remains an important driver of the solution. Since each large facility contributes 200 under-5 slots, building large facilities is an efficient way to satisfy both the total-slot and under-5 constraints simultaneously.

0.1.12 Conclusion

In summary, Model 2 shows that under more realistic planning assumptions, eliminating child care deserts in NYC requires a strategy centered on large-scale new construction, complemented by selective facility-level expansion. Compared with Model 1, the introduction of facility-specific expansion caps, piecewise increasing expansion costs, and same-zip distance restrictions substantially reduces the flexibility of expansion and makes feasible site selection a central component of the optimization.